Universal magnetotunnel conductance at a Weyl semimetal–layered Chern insulator junction

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We investigate electronic transport across a junction between a Weyl semimetal (WSM) and a layered Chern insulator (LCI) in the presence of a magnetic field perpendicular to the interface. The topological mismatch between the gapless Weyl semimetal and the momentum-resolved chiral edge modes of the layered Chern insulator leads to interface Fermi-arc states with a qualitatively distinct connectivity: unlike WSM–WSM junctions, the interface Fermi arcs are forced to reconnect through the Brillouin-zone boundary rather than terminating at the projections of the Weyl nodes. We analyze the spectrum and compute the magneto tunnel conductance mediated by the interface-localized states. We find that the conductance increases linearly with magnetic field at low fields and saturates beyond a critical field to a constant value that is independent of microscopic details such as interface coupling, arc geometry, and lattice-scale parameters. This universal saturation reflects a transport mechanism governed by the topological charge pumping associated with the Chern layers, rather than magnetic breakdown between Fermi arcs. We further show that, under specific conditions, a junction between two distinct Weyl semimetals can exhibit a similar saturation behavior, thereby clarifying the topological origin of the observed universality.

I. INTRODUCTION

Weyl semimetals (WSMs) host topologically protected band-touching points in the three-dimensional Brillouin zone, which act as monopoles of Berry curvature and give rise to a variety of unconventional electronic phenomena, including the chiral anomaly and anomalous magnetotransport responses [1–12]. A defining feature of WSMs is the presence of Fermi-arc surface states—open segments in the surface Brillouin zone that connect the projections of bulk Weyl nodes of opposite chirality[2, 3, 6, 7]. These states represent a striking manifestation of bulk–boundary correspondence and play a central role in transport phenomena unique to Weyl systems.

Interfaces between two Weyl semimetals provide a particularly rich setting for exploring the interplay between bulk topology, surface states, and transport. When the projections of the Weyl nodes on the two sides of the interface do not coincide, the interface generically hosts Fermi-arc states formed from the hybridization of surface arcs of the individual WSMs [13–19]. When a magnetic field is applied perpendicular to the interface, the interface Fermi arcs enable transport across the junction, with the current carried by the chiral zeroth Landau levels [20] being transmitted between the two WSMs. Depending on how the arcs connect the projected Weyl nodes, the resulting transport can exhibit perfect transmission or perfect reflection at low fields, followed by a saturation of the magnetotunnel conductance at high fields due to magnetic breakdown at close encounters of Fermi arcs [20–23]. In such WSM–WSM junctions, the value of the saturated conductance is controlled by microscopic details of the interface Fermi-arc geometry, such as the arc separation, intersection angle, and hybridization gap.

More recently, it has been shown that multiple close encounters between interface Fermi arcs can lead to non–Shubnikov–de Haas quantum oscillations in the magnetotunnel conductance, originating from momentum-space Aharonov–Bohm interference of semiclassical trajectories along the arcs [21]. These studies have firmly established interface Fermi arcs as active carriers of magnetotransport and have clarified the role of magnetic breakdown in determining the high-field response of WSM–WSM junctions.

In this work, we explore a qualitatively different interface geometry: a junction between a WSM and a layered Chern insulator (LCI). The setup is demonstrated in Figure 1. An LCI may be viewed as a stack of two-dimensional Chern insulators, characterized by a nonzero momentum-resolved Chern number and supporting chiral edge modes for each conserved momentum along the stacking direction. Unlike a Weyl semimetal, the LCI is fully gapped in the bulk and does not host Weyl nodes. The junction between a WSM and an LCI therefore represents a fundamentally distinct topological interface, where gapless Weyl fermions on one side couple to momentum-resolved chiral edge states on the other.

We show that this topological mismatch leads to interface Fermi-arc states with a connectivity that has no analog in WSM–WSM junctions. In particular, the interface Fermi arcs are not terminated by Weyl-node projections on both ends. Instead, they are forced to reconnect through the boundary of the Brillouin zone, reflecting the presence of a nontrivial Chern number on the LCI side. This connectivity is topologically enforced and persists independently of microscopic details such as interface coupling strength or lattice-scale parameters.

We investigate the consequences of this unconventional interface spectrum for magnetotransport across the junction. Focusing on a magnetic field perpendicular to the

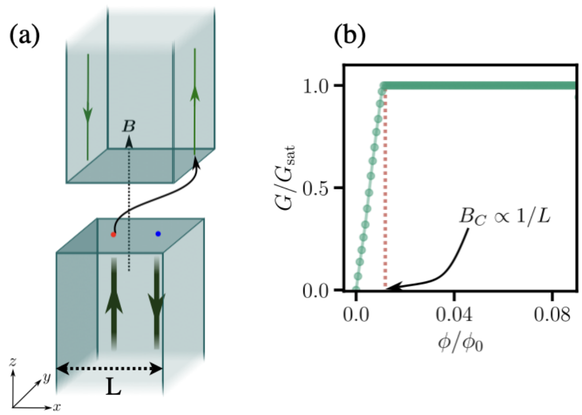

FIG. 1. (a) **Schematic diagram of the set up:** A junction of a WSM (bottom) and a LCI slab (top) is placed in a perpendicular magnetic field. In the WSM slab, currents carried by only the chiral Landau level states (green arrows) can transmits to LCI slab. The interface states (see Fig. 2) which are localized along z but are propagating in the plane of the interface redirects the current carried by the chiral LL states to the chiral edge states (green arrows on the y-z surfaces) of the LCI slab. (b) **Summary of the main result:** There are two regimes for magneto-tunnel conductance G- a linear regime which is followed by saturation to a universal value $G_{sat} = \frac{e^2}{h} L_y$ which is independent of any microscopic details of the system. The tunnel conductance in the linear regime $G = \frac{e^3}{h^2} \mathbf{A} \cdot \mathbf{B}$, where A is the area of interface, is also independent of any microscopic details of the system.

interface, we compute the magnetotunnel conductance mediated by the interface-localized states. We find that, while the conductance increases linearly with magnetic field at low fields, it saturates beyond a critical field B_c to a constant value that is independent of microscopic details of the system, including the precise shape and length of the interface Fermi arcs (See Fig. 1 for the setup and the main result). We find that the critical field at which saturation occurs scales as $B_c \propto 1/L$. Thus, in the thermodynamic limit, the tunnel conductance saturates to a universal value even for a very small field. This behavior stands in sharp contrast to WSM–WSM junctions, where conductance saturation arises from magnetic breakdown and depends explicitly on interface geometry. In the WSM–LCI junction, the universal saturation instead reflects a transport mechanism governed by topological charge pumping associated with the Chern layers of the LCI.

Finally, we show that a similar saturation behavior can also arise in junctions between two distinct WSMs, provided specific conditions are met that effectively mimic the role of a momentum-resolved Chern insulator. This comparison clarifies the topological origin of the universality observed in the WSM–LCI junction and highlights the essential role played by the Chern character of the interface states.

The remainder of this paper is organized as follows. In Sec. II, we introduce the model for the WSM–LCI junction and analyze the interface-localized states. In Sec. III, we compute the magnetotunnel conductance and demonstrate the emergence of universal saturation at high magnetic fields. In Sec. IV, we discuss the relation to WSM–WSM junctions and identify the conditions under which similar behavior can arise. We conclude in Sec. V with a summary and outlook.

II. INTERFACE STATES FOR WSM-LCI JUNCTION.

A. Model

We begin from a minimal two-band lattice model on a cubic lattice that realizes a Weyl semimetal phase with two Weyl nodes separated along the k_x direction. The Bloch Hamiltonian is written as

$$H_{\rm WSM}(\mathbf{k}) = M(\mathbf{k})\sigma_x + v_y \sin k_y a \sigma_y + v_z \sin k_z a \sigma_z \quad (1)$$

with $M(k) = \frac{v_x}{\sin k_0 a}(2 + m - \cos k_x a - \cos k_y a - \cos k_z a)$, where $\sigma_{x,y,z}$ are Pauli matrices acting on the pseudospin space, and a is the lattice constant which we set to unity. For an appropriate range of the parameter m, the bulk spectrum hosts two Weyl nodes located at $\mathbf{k} = (\pm k_0, 0, 0)$, with $k_0 = \cos^{-1}(m)$. In what follows, we set $k_0 = \pi/2$ so that the two Weyl nodes are at maximum separation. These nodes carry opposite chirality and give rise to Fermi-arc surface states on boundaries perpendicular to the z and y directions. In $H_{\rm WSM}$, the v_i $(i = x, y, z)$ are the velocity components of the Weyl fermion along the i'th direction and here we define a ratio $v_r = v_y/v_x$ which plays an important role in transport and will be discussed later.

The LCI phase is obtained from the same parent Hamiltonian by continuously tuning parameters such that the two Weyl nodes move towards the boundary of the Brillouin zone along k_x and annihilate pairwise. After annihilation, the bulk spectrum becomes fully gapped. However, for each fixed value of momentum along the k_x direction, the effective two-dimensional subsystem carries a nonzero Chern number. The resulting phase can therefore be viewed as a stack of two-dimensional Chern insulators and supports chiral boundary modes for each momentum in the topological regime. In this sense, the LCI is the gapped descendant of the Weyl semimetal obtained via Weyl node annihilation at the BZ boundary.

To form the junction, we consider a slab geometry along the z-direction and place the interface at $z = 0$. The WSM occupies $z < 0$, while the layered Chern insulator occupies $z > 0$. The translation symmetry along the x and y directions is preserved and (k_x, k_y) remain good quantum numbers. The interface Brillouin zone (IBZ) is, therefore, spanned by (k_x, k_y).

Prior to forming the junction, we rotate the LCI slab by 90° about the z-axis. This choice serves two purposes. First, it ensures that the underlying cubic lattices of the

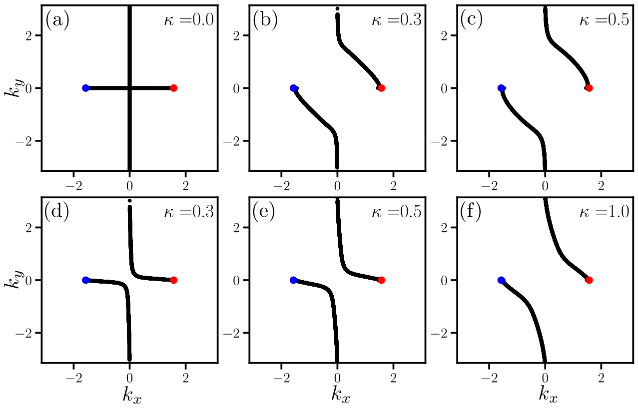

FIG. 2. Shows interface localized states of the WSM-LCI junction for a few values of tunneling coupling κ. (a) When $\kappa = 0$, the interface hosts the Fermi arc surface states from the WSM side and the chiral egde states from the LCI side. At finite coupling, they hybridize. The interface-localized states form continuous open contour that connects the projected Weyl nodes on the WSM side through the boundary of the the interface BZ. The first and second row are for two different values of the bulk parameter $v_r = 0.2$ and 1 respectively.

two slabs remain aligned at the interface, so that no additional superlattice structure is required. For a generic rotation angle, the two lattices would become incommensurate, necessitating the construction of a supercell to define a common interface Brillouin zone. Such a superlattice treatment would substantially increase the computational cost without introducing qualitatively new physics. Second, the 90° rotation ensures that, at zero interface coupling, the Fermi arcs of the WSM and the chiral edge modes of the LCI intersect at right angles in the IBZ. This orthogonal crossing provides a transparent starting point for analyzing the reconstruction of interface states upon hybridization. The 90° rotation also ensures that the transverse surface states which are on $\hat{x}$-$\hat{z}$ plane on the WSM are on the $\hat{y}$-$\hat{z}$ plane on the LCI and are not connected.

B. Interface Coupling

The full Hamiltonian of junction is

$$H = H_{\text{WSM}} + H_{\text{LCI}} + H_{\text{int}} \tag{2}$$

where H_{WSM} and H_{LCI} describe the two subsystems in slab geometry along z. The interface coupling connects the top layer of the WSM to the bottom layer of the LCI and is parametrized by two real parameters κ and u

$$H_{\text{int}} = \kappa \sum_{\mathbf{k}_\parallel} \left(c^\dagger_{\text{WSM},z=0}(\mathbf{k}_\parallel)(u\sigma_x + i\sigma_z)c_{\text{LCI},z=0}(\mathbf{k}_\parallel) + H.c \right) \tag{3}$$

with $\mathbf{k}_\parallel = (k_x, k_y)$. In general the coupling matrix, can also include a term proportional to σ_y. As previously discussed in Ref. 16, such a term would not have effect on the Fermi arc reconstruction at the interface because the surface states localized at the interface of the top/bottom slabs are up/down pseudospin polarized along the y direction in the Pauli space and hence a σ_y term cannot contribute in hybridization. The parameter u indicates an imbalance between pseudospin-preserving and pseudospin-flipping hoppings across the interface in contrast to the bulk hopping. In what follows we will fix $u = 0.5$ and use κ as a control parameter which measures the strength of hybridization across the interface. In the decoupled limit $\kappa = 0$, the interface localized states consist of the Fermi arc states of the WSM and the chiral edge states of LCI slabs. For finite κ, the interface states are obtained from exact diagonalization of the full coupled Hamiltonian for each (k_x, k_y).

C. Interface localized states

To determine the interface spectrum, we numerically diagonalize the slab Hamiltonian at fixed (k_x, k_y) and identify states localized near $z = 0$ via their spatial probability distribution. The resulting dispersion $E(k_x, k_y)$ reveals interface-localized bands lying within the bulk gap of the LCI and within the projected Weyl node region of the WSM.

At $\kappa = 0$, the IBZ contains two distinct sets of gapless states: (i) Fermi arcs from the WSM surface connecting the projections of the Weyl nodes separated along k_x, and (ii) chiral edge modes from the LCI which disperse along k_x. These states cross orthogonally without hybridization.

Upon turning on a finite interface coupling κ, hybridization at the crossings reconstructs the spectrum. We find that the resulting interface-localized states form continuous open contours that connect the projected Weyl nodes on the WSM side through the boundary of the BZ (see Fig. 2). Because the LCI does not host Weyl nodes, the interface Fermi arcs cannot terminate on node projections on both ends. Instead, they are forced to reconnect via the IBZ boundary, reflecting the nontrivial Chern number inherited from the Weyl-node annihilation process.

This connectivity is robust against variations of κ and other microscopic details, provided the bulk topology of the two subsystems remains unchanged. As we show in the following section, this distinctive Fermi-arc structure plays a central role in determining the magnetotransport properties of the WSM–LCI junction.

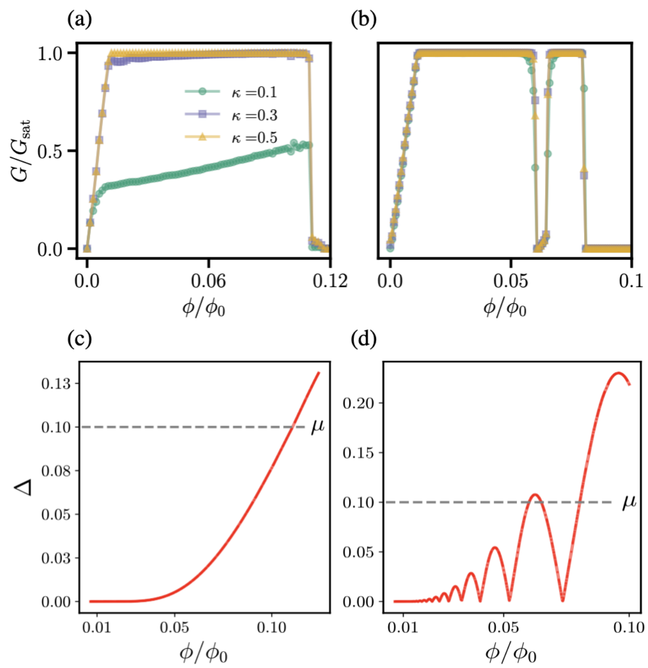

FIG. 3. (a) and (b) show magneto-tunnel conductance across the WSM-LCI junction for several values of interface coupling κ as a function of flux through the interface. In (a) the bulk parameter $v_r = v_y/v_x$=1.0 and in (b) it is 0.2. The system size along the x and y direction are $L_x = L_y = 100$. The chemical potential (measured relative to the energy of the Weyl point) is $\mu = 0.1$. Unless interface coupling parameter κ is small (see main text for an explanation), the conductance saturates to a universal value $G_{sat} = \frac{e^2}{h} L_y$ at flux $\phi/\phi_0 = 1/L_x$. (c) and (d) show the bulk gap Δ in the WSM induced by the magnetic field for $v_r = 1.0$ and 0.2 respectively. Whenever the field induced bulk gap Δ is larger than the chemical potential, the tunnel conductance drops to zero.

III. MAGNETO-TUNNEL CONDUCTANCE ACROSS JUNCTION

A. Zero-field transport

We first discuss transport in the absence of a magnetic field. Since the LCI is fully gapped in the bulk, only its surface states can participate in low-energy transport. On the WSM side, surface Fermi arcs exist on boundaries ($y = 0, L_y$) perpendicular to the z-direction. In the slab geometry considered here, current injected from the WSM side is carried by these surface states.

At zero field, bulk states in the WSM do not contribute to transmission across the junction at low energy, as there is no direct bulk channel in the LCI to receive them. Instead, transport proceeds entirely through interface-localized states that are exponentially localized in the z-direction but propagate in the (x, y)-plane. As discussed in our previous work in Ref. [22], the current arriving at the interface from the WSM surface is redirected along these interface states and subsequently transferred to the chiral edge modes of the LCI slab.

Thus, at $B = 0$, the conductance across the junction is determined solely by the overlap and hybridization between surface Fermi arcs of the WSM and chiral edge modes of the LCI. The bulk topology of the WSM does not directly contribute to transport in this limit.

B. Transport in perpendicular magnetic field

The situation changes dramatically in the presence of a magnetic field $\mathbf{B}$ parallel to $\hat{z}$ and perpendicular to the interface. The field quantizes the bulk states of the WSM to form Landau levels. In particular, each Weyl node generates a chiral zeroth Landau level (LL) that disperses linearly along the field direction. These chiral LLs provide bulk channels that carry current toward the interface.

In contrast, the LCI remains fully gapped in the bulk even in the presence of the magnetic field. Its low-energy transport continues to be carried by chiral edge modes associated with the momentum-resolved Chern number.

The essential mechanism for magnetotransport across the junction is therefore the following: current carried by the chiral Landau level in the WSM is redirected at the interface into chiral edge modes of the LCI. As discussed in previous works [20–22], this redirection occurs via the interface-localized states identified in Sec. II. The interface states act as a momentum-space bridge between the bulk chiral LL of the WSM and the boundary modes of the LCI.

In the presence of the magnetic field, scattering processes at the interface become possible. In particular, an electron propagating along one Fermi arc may tunnel to another arc and be reflected back into the WSM. Such processes correspond to magnetic breakdown between interface states and suppress the conductance by an exponentially small factor $(1 - e^{-B_0/B})$, where $B_0 \propto q_{FAs}^2$ and q_{FAs} denotes the momentum space distance between the two Fermi arcs at the interface, as shown earlier in Refs. 20 and 21.

As shown below, there exists a characteristic field B_c, determined solely by the system size perpendicular to the transport direction. When $B_c \lesssim B_0$, the effect of such scattering processes becomes negligible, and the tunnel conductance saturates to a constant value independent of microscopic details such as the interface coupling strength κ, the precise geometry of the Fermi arcs, or lattice-scale band parameters.

C. Field dependence and saturation of conductance

The magnetic field introduces a large degeneracy of chiral LL modes. For a system of dimensions $L_x \times L_y$, the number of chiral LL channels in the WSM is

$$N_{\rm LL} = B L_x L_y/\phi_0, \tag{4}$$

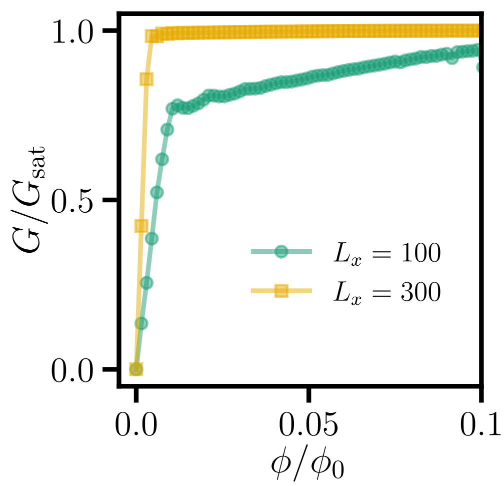

FIG. 4. Shows the system-size dependence of magneto-tunnel conductance across the WSM-LCI junction. For small tunnel coupling $\kappa = 0.2$, the scattering between the Fermi arcs at the interface is significant and hence the conductance does not saturate to universal value $G_{\rm sat} = \frac{e^2}{h} L_y$ for small system size along x direction. However for larger system size $L_x = 300$, the conductance saturates to the universal value G_{sat} at a flux value $\phi/\phi_0 = 1/L_x$. The system size along the y direction is fixed to $L_y = 50$.

reflecting the Landau degeneracy proportional to the magnetic flux (in units of flux quantum $\phi_0 = h/e$) through the interface plane.

On the LCI side, however, the number of available chiral edge modes is determined solely by the Chern character of the layered insulator and by the system size perpendicular to the direction of propagation. In the present geometry, the number of LCI chiral edge modes $N_{\rm edge} = L_y$ (here, the lattice constant has been set to unity), and is independent of the magnetic field.

This mismatch in channel scaling plays a central role in determining the magnetotunnel conductance. For weak magnetic fields, the number of incoming chiral LL modes is smaller than or comparable to the number of available LCI edge channels. In this regime, the conductance increases approximately linearly with B, reflecting the increasing number of transmitting chiral LL modes.

As the magnetic field is increased further, the number of WSM chiral LL modes eventually exceeds the number of available LCI edge modes. Beyond this point, additional LL channels cannot be accommodated by independent outgoing edge modes on the LCI side, and hence the transport becomes limited by the number of available LCI edge channels. The conductance therefore saturates to a constant value determined solely by the number of LCI chiral edge modes

$$G_{sat} = \frac{e^2}{h} L_y, \tag{5}$$

and becomes independent of further increases in the magnetic field. This behavior of the tunnel conductance as a function of magnetic field (computed by employing KWANT simmulation [24]) is demonstrated in Fig. 3.

The crossover field B_c at which saturation sets in is obtained by equating the number of chiral LL modes to the number of edge channels $B_c L_x L_y/\phi_0 = L_y$, resulting in $B_c = \phi_0/L_x$. Thus, the onset of saturation shifts to lower magnetic fields for wider systems. Importantly, the maximum conductance is independent of microscopic details such as the interface coupling parameter κ, the precise geometry of the Fermi arcs, or lattice-scale band parameters. Once the magnetic field is sufficiently strong, transport is controlled entirely by the topological constraint imposed by the finite number of LCI chiral edge modes. The universality of the saturation therefore reflects the Chern character of the LCI rather than details of interface scattering.

We note that the conductance for small interface coupling, as shown in Fig. 3(b), does not saturate to the constant value G_{sat}. This is because the momentum-space distance q_{FA} between the two Fermi arcs at the interface is small, due to the stronger scattering when B_0 is small compared to B_c. Nevertheless, for any thermodynamically large system, the saturation field $B_c \to 0$ (due to the large degeneracy of the chiral LLs), and hence the tunnel conductance should saturate to G_{sat} irrespective of the strength of the interface coupling parameter. This is demonstrated in Fig. 4 by plotting the tunnel conductance for larger system sizes.

We emphasize that Fermi arc states of the WSM, propagating towards the interface, also contribute to transport. However, in the tunneling conductance shown in Fig. 3, this contribution has been neglected. As demonstrated in our previous work [22], these states affect only the linear component of the tunneling conductance and does not modify the overall behavior discussed here. Additionally, for large system size in the thermodynamic limit, this contribution is negligible.

We notice that magneto tunnel conductance, shown in Fig. 3, behaves quite differently for the two values of $v_r = v_y/v_x = 1.0$ and 0.2. This is somewhat surprising since the naive expectation would be for the tunnel conductance to be qualitatively similar when bulk parameters such as the velocities v_x and v_y of the Weyl fermion are changed. In particular, we note that for $v_r = 0.2 < 1$, the conductance oscillates between saturation values and zero as a function of the magnetic field. This unusual behavior of the magneto-tunnel conductance can be seen to be linked to subtle changes in the bulk band structure at low energies of the WSM due to the magnetic field. Recent work have both theoretically established [25–32] as well as experimentally observed [33, 34] that an orbital magnetic field which is perpendicular to the direction of separation of Weyl nodes of opposite chirality couple them and a finite gap Δ opens at the Weyl point. Consequently, the chiral LL becomes gapped at low energies. Therefore, whenever the chemical potential lies in the bulk gap, the conductance drops to zero because there are no states to carry the current. For

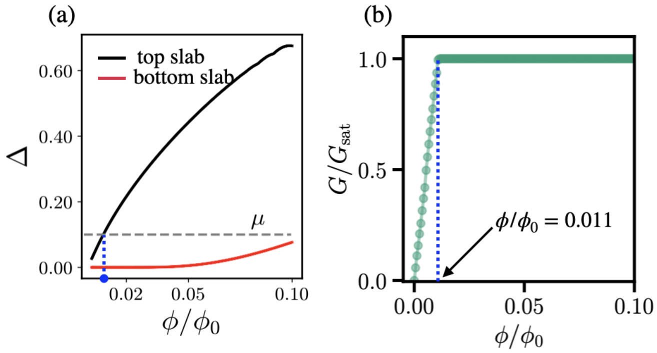

FIG. 5. (a) Magnetic field induced gap in the bulk of the WSM slabs and (b) the universal magneto-tunnel conductance across the junction. For the junction, we adopt the model introduced in Eq. 1 of our previous work [22] with momentum space separation between the nodes $Q = 2k_0 = \pi$ and 5.6 for bottom and top slabs respectively. We assumed tunnel coupling strength $\kappa = 1.0$ and the system size along x, y directions to be $L_x = 100$ and $L_y = 50$ respectively. For a chemical potential $\mu = 0.1$, the junction effectively behaves like a WSM-LCI junction for flux values beyond $\phi/\phi_0 \approx 0.011$ (marked by a blue dot in (a)). (b) As expected, the conductance saturates to universal value G_{sat} at the corresponding flux value.

$v_r = 1.0 \geq 1$, the induced gap Δ monotonically increases with the field and for $v_r = 0.2 < 1$ it oscillates with field (see Figs. 3). Hence, by comparing 3(a) and (c), and (b) and (d), we see that whenever the chemical potential is below the gap, the conductance drops to zero.

IV. UNIVERSAL CONDUCTANCE FOR WSM-WSM JUNCTION

In the previous section we showed that the magnetotunnel conductance across a WSM–LCI junction saturates to a universal value at sufficiently large magnetic fields. In this section we demonstrate that a similar universal behavior can also arise in a junction formed between two WSMs under appropriate conditions.

As we discussed in the previous section, an external magnetic field which is applied along a direction perpendicular to the separation of a pair of Weyl nodes in momentum space can couple them and gap them pairwise. The resulting insulating state could be topologically nontrivial. As shown in recent works [29–31], the Landau quantization can lead to the formation of either a normal insulating state or a layered Chern insulator depending on the separation between the Weyl nodes. For sufficiently large momentum-space separation between the nodes, the system effectively realizes a stack of two-dimensional Chern insulators. In this regime the magnetic field drives the system into a LCI phase where the Fermi arc states of the WSM evolve to the chiral edge states of LCI.

Motivated by this observation, we consider a junction between two WSMs whose Weyl node separations are chosen differently. For the junction, we adopt the model introduced in Eq. 1 of our previous work [22], with the following modification: In the top WSM, the momentum-space separation between the Weyl nodes is chosen to be sufficiently large so that, in the presence of a perpendicular magnetic field, the system effectively behaves as a layered Chern insulator. In contrast, in the bottom WSM is chosen to have a smaller node separation such that the magnetic-field–induced gap remains small and the chemical potential can be tuned so that the chiral zeroth Landau level persists in the bulk and continues to participate in transport

Under these conditions, once the magnetic field exceeds a certain threshold value, the junction effectively behaves as a WSM–LCI interface. The bulk states of the top WSM reorganize into momentum-resolved Chern insulating layers that support chiral edge modes, while the bottom WSM continues to host chiral Landau-level channels that carry current toward the interface. The transport mechanism therefore becomes identical to the one discussed in Sec. III: current carried by the chiral Landau level of the WSM is redirected at the interface into the chiral edge modes of the effective LCI phase.

As a consequence, the magnetotunnel conductance of the WSM–WSM junction exhibits the same qualitative behavior found for the WSM–LCI system (see Fig. 5). At low magnetic fields, the conductance increases with field due to the increasing degeneracy of the chiral Landau-level channels. Beyond a critical magnetic field B_c, however, the number of available outgoing chiral edge modes on the effective LCI side becomes the limiting factor for transport. The conductance therefore saturates to a constant value that is independent of further increases in magnetic field.

Importantly, the saturated conductance is insensitive to microscopic details such as the precise separation of Weyl nodes, the interface coupling strength, or other band-structure parameters. Instead, it is determined solely by the number of chiral edge modes supported by the effective layered Chern insulator. The emergence of this universal saturation thus reflects the topological nature of the interface transport.

This analysis shows that the universal conductance plateau identified in Sec. III is not restricted to junctions involving a genuine layered Chern insulator. Rather, it can also appear in appropriately tuned WSM–WSM junctions where one of the WSMs effectively realizes a layered Chern insulating phase in the presence of a magnetic field.

V. DISCUSSION AND CONCLUSION

In this work we investigated interface states and magnetotransport across a junction between a WSM and a LCI. Using a minimal two-band lattice model, we constructed the junction in a slab geometry and determined the interface-localized states via exact diagonalization. We found that the hybridization between the surface Fermi arcs of the WSM and the chiral edge modes of the LCI produces reconstructed interface states with a distinct connectivity in the interface Brillouin zone. In particular, since the LCI does not host Weyl nodes, the interface Fermi arcs cannot terminate on the node projection on the LCI side of the interface and instead reconnect through the boundary of the Brillouin zone, reflecting the Chern character of the LCI inherited from the Weyl-node annihilation process.

We then studied magnetotunnel transport across the junction in the presence of a magnetic field perpendicular to the interface. While transport at zero field is mediated only by surface states, a magnetic field introduces chiral Landau-level channels in the bulk of the WSM that carry current toward the interface. These modes are redirected through the interface states into the chiral edge modes of the LCI. Because the number of incoming chiral Landau-level modes increases with magnetic field whereas the number of available edge channels in the LCI remain fixed, the magnetoconductance increases at low fields but eventually saturates to a constant value. The crossover field scales as $B_c \propto 1/L$, and the saturated conductance depends only on the number of LCI edge modes, making it independent of microscopic details of the interface.

We further showed that similar universal conductance saturation can also arise in junctions between two WSMs when a magnetic field effectively drives one of them into a LCI regime.

Here we briefly comment on the robustness of the Fermi-arc–mediated magnetotransport across the interface in the presence of disorder. Interface disorder introduces a finite lifetime τ_{life}, which affects transport by enabling scattering between Fermi-arc states. As shown recently in Ref. 23, this leads to the emergence of a characteristic field scale B_0^*, defined by the condition that the Fermi-arc dwell time becomes comparable to τ_{life}.

In the high-field regime ($B \gg B_0^*$), the dwell time is short and inter-arc scattering is ineffective, so the tunnel magnetoconductance coincides with the clean-interface result and is essentially insensitive to disorder. In contrast, in the low-field regime ($B \ll B_0^*$), disorder can modify the slope of the linear magnetoconductance due to enhanced scattering and partial equilibration between transport channels.

Importantly, the universal conductance saturation at large magnetic fields remains robust against disorder, since it is ultimately determined by the number of available chiral edge modes in the layered Chern insulator, which are topologically protected and insensitive to weak interface scattering.

Experimentally, the predicted conductance plateau could be probed in heterostructures combining WSMs with layered quantum anomalous Hall systems or magnetic topological insulator multilayers [3, 35, 36], or in engineered WSM–WSM interfaces [30, 31] tuned to the appropriate regime.

Overall, our results highlight how the interplay between chiral Landau levels and momentum-resolved Chern edge modes at topological interfaces can produce robust and universal transport behavior controlled primarily by topology rather than microscopic bandstructure details.

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